



# The hidden order of Turing patterns in arid and semi-arid vegetation ecosystems

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Vegetation Turing patterns play a critical role in the ecological functioning of arid and semi-arid ecosystems. However, the long-range spatial features of these patterns have been neglected compared to short-range features like patch shape and spatial wavelength. Drawing inspiration from hyperuniform structures in material science, we find that the arid and semi-arid vegetation Turing pattern exhibits long-range dispersion similar to hyperuniformity. As the degree of hyperuniformity of the vegetation Turing pattern increases, so does the water-use efficiency of the vegetation. This finding supports previous studies that suggest that Turing patterns represent a spatially optimized self-organization of ecosystems for water acquisition. The degree of hyperuniformity of Turing-type ecosystems exhibits significant critical slowing down near the tipping point, indicating that these ecosystems have non-negligible transient dynamical behavior. Reduced rainfall not only decreases the resilience of the steady state of the ecosystem but also slows down the rate of spatial optimization of water-use efficiency in long transient regimes. We propose that the degree of hyperuniformity indicates the spatial resilience of Turing-type ecosystems after strong, short-term disturbances. Spatially heterogeneous disturbances that reduce hyperuniformity lead to longer recovery times than spatially homogeneous disturbances that maintain hyperuniformity.

Turing pattern | hyperuniformity | transient dynamics | critical slowing down | disturbance

Spatial self-organization of vegetation patterns, such as spot, labyrinth, and gap, is commonly observed in arid and semi-arid ecosystems (Fig. 1 A1–A4) (1–3). These patterns have been explained through many mathematical models that account for the mechanisms of plant–water interaction (4–7). Despite differences in form, these models are consistent with the activator–inhibitor principle that Turing first proposed in 1952 (8), which is also known as the scale-dependent feedback principle in ecology (Fig. 1B) (9–11). In addition to the primary Turing instability, these scale-dependent feedback models were found to have multiple instabilities that are closely related to real-world vegetation patterns (12, 13), thus serving as a theoretical basis for the spatial self-organization of vegetation patterns in resource-limited environments (14–16). Specifically, plants promote the growth of neighboring plants by extracting water from the surrounding environment, resulting in short-range positive feedback that leads to aggregation in the scale-dependent feedback (17, 18). Conversely, plants create zones of water depletion that inhibit the growth of distant plants due to limited water availability, leading to long-range negative feedback that leads to regularity in the scale-dependent feedback (9). Aggregation in arid and semi-arid ecosystems is typically visually identified through the clustering of plants, which form distinct vegetation patches that stand out from the surrounding bare ground (19). This characteristic is considered a local feature. On the other hand, regularity refers to the periodic distribution of vegetation patches in a spatial pattern (20). Regularity is considered a short-range feature because the spatial wavelength used to characterize regularity measures the average distance between vegetation patches (21).

The Turing pattern actually exhibits a hidden order known as long-range dispersion, which implies that it is distributed more uniformly than the spatially random distribution at larger spatial scales (22). The term “hidden” implies that long-range dispersion is difficult to detect visually [interested readers can refer to figure 3 of Torquato (23)]. Interestingly, studies have confirmed that the spatial distribution of termite mounds in semi-arid ecosystems is more uniform than the spatial Poisson distribution (24–26). However, these studies have not proposed a metric to quantify the degree of deviation from the spatial Poisson distribution. Recently, a metric to describe anomalous material structure has been discovered in materials science, leading to the discovery of a large class of novel spatial structures called hyperuniform states (23, 27). Density fluctuations (*Methods*) in a hyperuniform state (when normalized by those for typical disordered systems) vanish entirely at infinitely large length scales. Initially, the degree of hyperuniformity provides an order

## Significance

In arid and semi-arid climates, the self-organization of plants forms striking Turing patterns. However, the long-range hyperuniformity of these patterns has long been overlooked. The degree of hyperuniformity of the Turing pattern is positively correlated with water-use efficiency by vegetation. Thus, Turing patterns represent a spatially optimized self-organization structure for water acquisition. The hyperuniformity reveals the critical slowing down of spatial evolution when rainfall approaches the tipping point. Decreasing rainfall can lead Turing-type arid and semi-arid ecosystems to long-term transient regimes with low resilience. Disruption of hyperuniform structures causes severe damage to Turing-type ecosystem resilience. By understanding hyperuniformity of Turing-type ecosystems, we can develop effective strategies for mitigating the impacts of disturbances and enhancing the sustainability of arid and semi-arid ecosystems.

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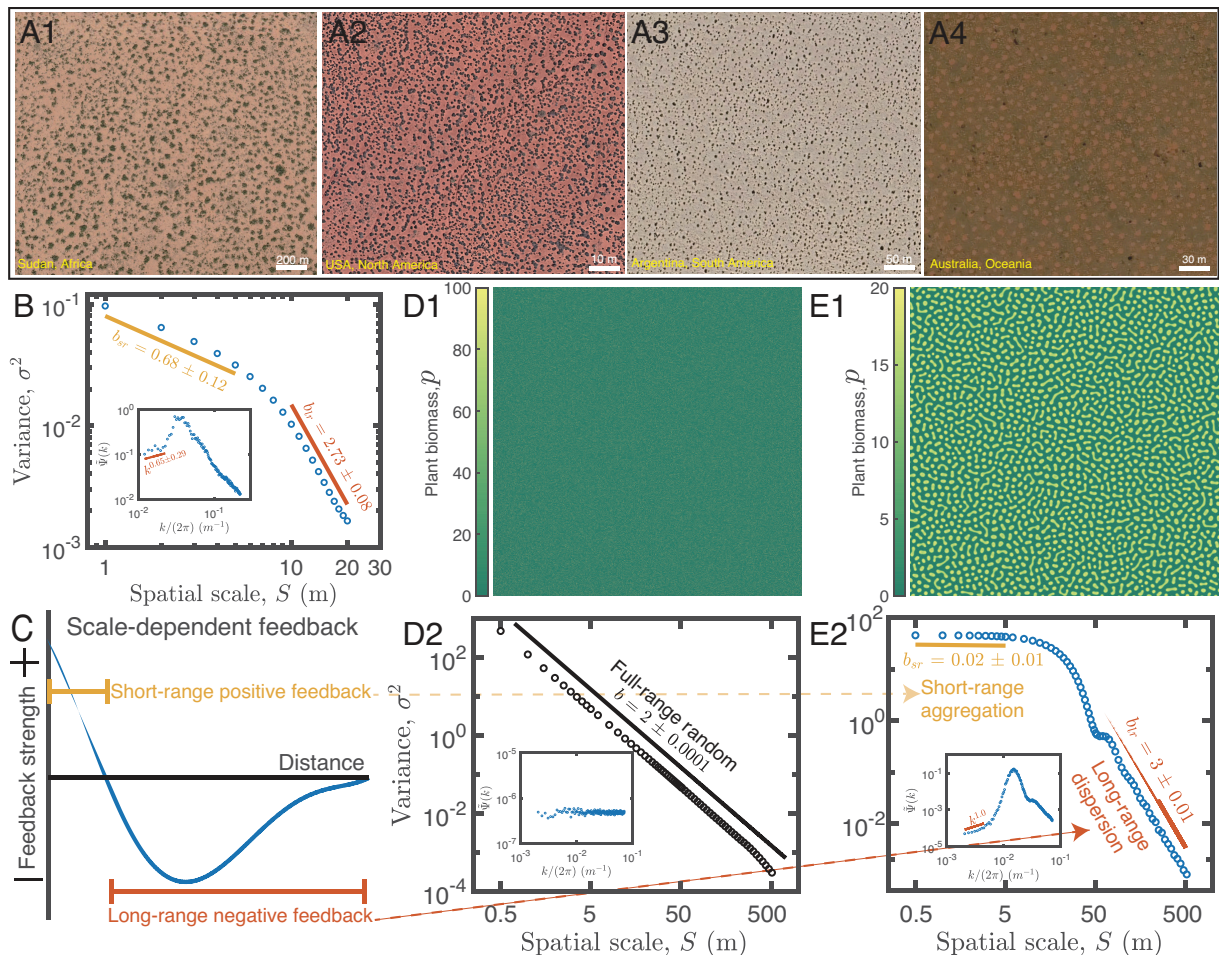
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**Fig. 1.** Density fluctuation reveals the long-range hyperuniform state of arid and semi-arid vegetation ecosystems. (A1–A4) Diverse vegetation Turing patterns across the world. (A1) A spot pattern in Sudan (Location: 11.4461°N, 27.8938°E); (2) Mixtures of spot and labyrinth patterns in the United States (Location: 38.1738°N, 109.7099°W); (3) A spot pattern in Argentina (Location: 23.4363°S, 65.9768°W); (4) A gap pattern (also called by fairy circle) in Australia (Location: 23.4484°S, 119.8529°E). These images are from Google Earth Pro. (B) The density fluctuation and circularly averaged spectral density (*Methods*) of the vegetation pattern in A4. The corresponding results of A1–A3 are shown in *SI Appendix, Fig. S1*. (C) Schematic representation of the scale-dependent feedback. Short-range positive feedback means that plants facilitate the growth of other plants nearby by enhancing water acquisition. Long-range negative feedback means that plants inhibit the growth of plants far away due to limited water. (D1) The initial vegetation condition has a spatially full-range random configuration. (E1) The steady Turing pattern developed from D1 according to a scale-dependent feedback model (Eq. 1). (D2 and E2) The density fluctuation and circularly averaged spectral density (*Insets*) of the spatial patterns (D1 and E1), respectively. Short-range positive feedback leads to short-range aggregation of plants, that is, vegetation patches. Long-range negative feedback leads to a long-range dispersion configuration of vegetation patches, a hyperuniform state. The diachronic changes in two areas in Niger show the development of the hyperuniform state of a vegetation Turing pattern (*SI Appendix, Fig. S2*).

metric for the degree to which large-scale fluctuations are suppressed in the point pattern (27). Subsequently, hyperuniformity has been extended to treat fluctuations in point patterns to those associated with scalar fields, and it can also characterize small-scale spatial features (28, 29). Thus, we propose density fluctuation as a spatial metric for measuring aggregation or dispersion of Turing patterns in arid and semi-arid vegetation across spatial scales.

In addition to the striking geometry of Turing patterns, understanding the emergent properties of vegetation Turing patterns is critical to the functioning of arid and semi-arid ecosystems (30–32). The aggregation of plants has been explained as a resource concentration mechanism in which plants concentrate water from their surroundings to support their survival (33). On the other hand, many studies suggest that regular vegetation patterns optimize water-use efficiency in arid and semi-arid climates (5, 6, 17). Vegetation patches modify the environment to reallocate limited water, thereby increasing productivity by improving water-use efficiency (31). However, no study has proposed a method to evaluate the water-use efficiency of arid and semi-arid vegetation Turing patterns. Water is usually a variable in mathematical models of scale-dependent feedback. Water levels can be obtained

through direct numerical simulation, assuming that water is of fixed density and incompressible. Then, water flow paths can be extracted from water levels based on existing hydrological indicators (34, 35). The water-use efficiency of vegetation patches can be assessed based on the length of the water flow path. As a result, the relationship between water-use efficiency and long-range dispersion of the pattern can be directly quantified.

Another critical function of vegetation Turing patterns in arid and semi-arid ecosystems is to indicate ecosystem collapse (36, 37). Mathematical models of scale-dependent feedback and field evidence indicate that on a flat terrain, vegetation patches form gap, labyrinth, and spot patterns on a descending rainfall gradient (2, 37). The presence of spots is seen as an indicator of approaching collapse (36). While these models show that the mean biomass exhibits critical slowing down close to the tipping point (38), it is unclear whether the spatial features of the Turing patterns are affected by critical slowing down as the ecosystem approaches the tipping point. These models are high-dimensional systems with a tipping point (39). As such, it is difficult to imagine that the spatial features of the Turing patterns are unaffected by critical slowing down. Most previous studies have defaulted to Turing patterns of



arid and semi-arid vegetation in the asymptotic state because plants form vegetation patches quickly during the early stage of pattern formation in numerical simulations. Since the spatial evolution of the long-range dispersion of Turing patterns is ignored, the transient aspect of these patterns and their effect on the resilience of the ecosystem remains unknown.

Recent theoretical studies have advanced the understanding of the spatial resilience of vegetation Turing patterns in arid and semi-arid ecosystems (40). The Busse balloon theory proposes that a slow decline in rainfall resulting from climate warming can lead to the Eckhaus destabilization of the patterned vegetation state, ultimately allowing for the evasion of a tipping point (41, 42). Increased spatial heterogeneity of the aridity parameter can improve the spatial resilience of vegetation, thus preventing catastrophic vegetation collapse (43). These theoretical results indicate that the spatial resilience of these ecosystems depends not solely on rainfall but also on the spatial structure of these ecosystems (44). In addition to slow, long-term changes in rainfall (or aridity parameter), arid and semi-arid ecosystems also face short-term, drastic disturbances such as flash droughts, overgrazing, and wildfires (45, 46). Numerical disturbance experiments typically involve uniform biomass removal across a patterned state (38, 47–49). While these numerical experiments can effectively simulate flash droughts, they may not accurately capture the effects of overgrazing and wildfires. These disturbances often disrupt the spatial structure (50), which these numerical disturbance experiments do not consider. Therefore, it is essential to quantify the relationship between the spatial structure of vegetation and the spatial resilience under intense short-term disturbance to inform ecosystem management strategies.

In this study, we aim to investigate the evolution of Turing patterns in arid and semi-arid ecosystems, focusing on quantifying long-range dispersion rather than pattern formation. Specifically, we analyzed the degree of hyperuniformity in vegetation Turing patterns of four ecological models and showed that its evolution is linked to the spatial optimization of water-use efficiency. We also found that critical slowing down of the spatial evolution occurs as the ecosystem approaches the tipping point, leading to a long transient regime with low water-use efficiency and reduced system resilience. Furthermore, disturbances that disrupt spatial structure cause a significant reduction in spatial resilience. Therefore, we propose the degree of hyperuniformity as a metric for assessing spatial resilience after intense short-term disturbances.

## Results

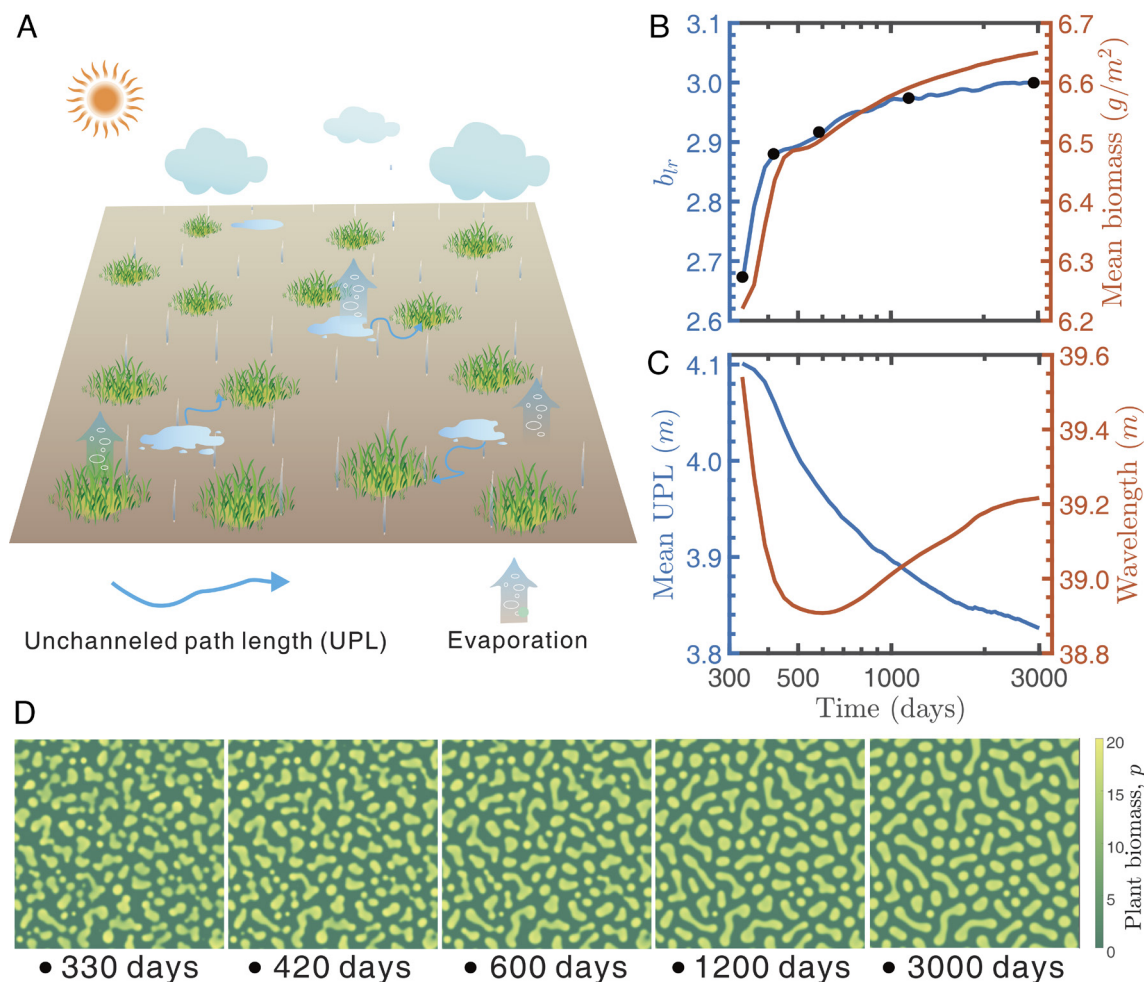
**Hyperuniform State and Concomitant Spatial Optimization.** Eq. 1 describes scale-dependent feedback between plants and water (the other models are in *SI Appendix, Text S1*). The nonlinear terms  $\alpha \frac{P + k_2 W_0}{P + k_2} O$  and  $\frac{g_{\max} W}{W + k_1} P$  demonstrate that plants can locally enhance the infiltration rate of surface water, and the increase in soil water content can support plant growth locally, thereby creating short-range positive feedback of “water infiltration–plant growth” (i.e., short-range facilitation). The diffusion terms of Eq. 1 implicitly encode long-range negative feedback. Vegetation creates patches locally through short-range positive feedback and generates a biomass gradient between vegetation patches and bare ground. Due to the substantial infiltration of surface water into the soil at vegetation patches, surface water generates a gradient between vegetation patches and bare ground opposite to biomass. As nonlocal negative feedback, diffusion tries to “eliminate” the gradient of biomass or water and, in effect, continuously transports surface water into the vegetation patch. Thus, with enough surface

water, vegetation can spread to the entire space and establish a spatially homogeneous state. When there is insufficient surface water, vegetation patches indirectly compete for water through diffusion processes, leading to the depletion of the soil water content of bare ground between vegetation patches (i.e., long-range competition). In this scenario, the expansion of each vegetation patch is restricted. As a result of short-range facilitation and long-range competition, a regular vegetation pattern emerges.

The natural and simulated Turing patterns in the arid and semi-arid ecosystems are characterized by short-range aggregation and long-range dispersion, as revealed by the density fluctuation analysis (Fig. 1 *B* and *E* and *SI Appendix, Figs. S1 and S3–S5*). The interplay between short-range facilitation and long-range competition, starting from the initial spatial Poisson distribution of vegetation in the numerical simulation, leads to aggregation at short spatial scales and dispersion at long spatial scales (Fig. 1 *C–E*). While an isolated vegetation patch reflects the aggregation feature, the spatial configuration of vegetation patches reflects the dispersion feature. The 40-y interval aerial photos from protected areas in southern Niger show that the formation of vegetation Turing patterns accompanied the natural generation of long-range dispersion features (*SI Appendix, Fig. S2*). The degree of long-range dispersion, quantified by the hyperuniform index  $b_{lr}$ , indicates that the Turing pattern of the arid and semi-arid ecosystems is hyperuniform. The evolution of the spatial structure toward the hyperuniform upper limit ( $b_{lr} = 3$ ) is accompanied by a decreasing mean unchanneled path length (UPL, *Methods*), which shows that surface water on the bare ground can flow over shorter distances to reach vegetated patches (Fig. 2 *B* and *C* and *SI Appendix, Figs. S3–S5*). Thus, the evolution of the hyperuniform state of the Turing pattern is associated with optimizing the water-use efficiency of the vegetation. However, the nonmonotonic variation of the wavelength can only characterize the spatial periodicity of the pattern and does not correlate with the spatial optimization of water-use efficiency (Fig. 2 *B* and *C*).

**Critical Slowing Down Leading to Non-Negligible Transient Dynamics.** We conducted a numerical examination of the behavior of  $b_{lr}$  in the parameter space of rainfall. For steady states, the mean biomass linearly decreases with rainfall in the parameter space of the Turing pattern (Fig. 3*A* and *SI Appendix, Fig. S6A*), and spatial patterns of gap, labyrinth, and spot emerge in sequence (Fig. 3*C*). For gap and labyrinth patterns, the  $b_{lr}$  of steady states gradually approaches the upper limit 3 with decreasing rainfall (Fig. 3*B*). When spots emerge, the  $b_{lr}$  of steady states always converges to the upper limit 3 with decreasing rainfall until the tipping point (Fig. 3 *B–D* and *SI Appendix, Fig. S6*). Notably, the time taken for  $b_{lr}$  to reach 3 increases exponentially with decreasing rainfall (Fig. 3*B* and *SI Appendix, Fig. S6B*). As the ecosystem approaches the tipping point, the temporal changes in mean biomass and  $b_{lr}$  reveal that decreasing rainfall causes the vegetation to remain in a state of a loss of hyperuniformity (i.e.,  $b_{lr} < 2$ ), resulting in low biomass for an extended period. After the ecosystem evolves to achieve hyperuniformity (i.e.,  $b_{lr} > 2$ ), biomass rapidly increases (Fig. 3 *E* and *F*). It is important to note that the evolution of  $b_{lr}$  is accompanied by the spatial optimization of water acquisition by vegetation. Therefore, decreasing rainfall reduces the mean biomass in steady states and significantly slows down the spatial optimization of water-use efficiency by vegetation in long transient regimes.

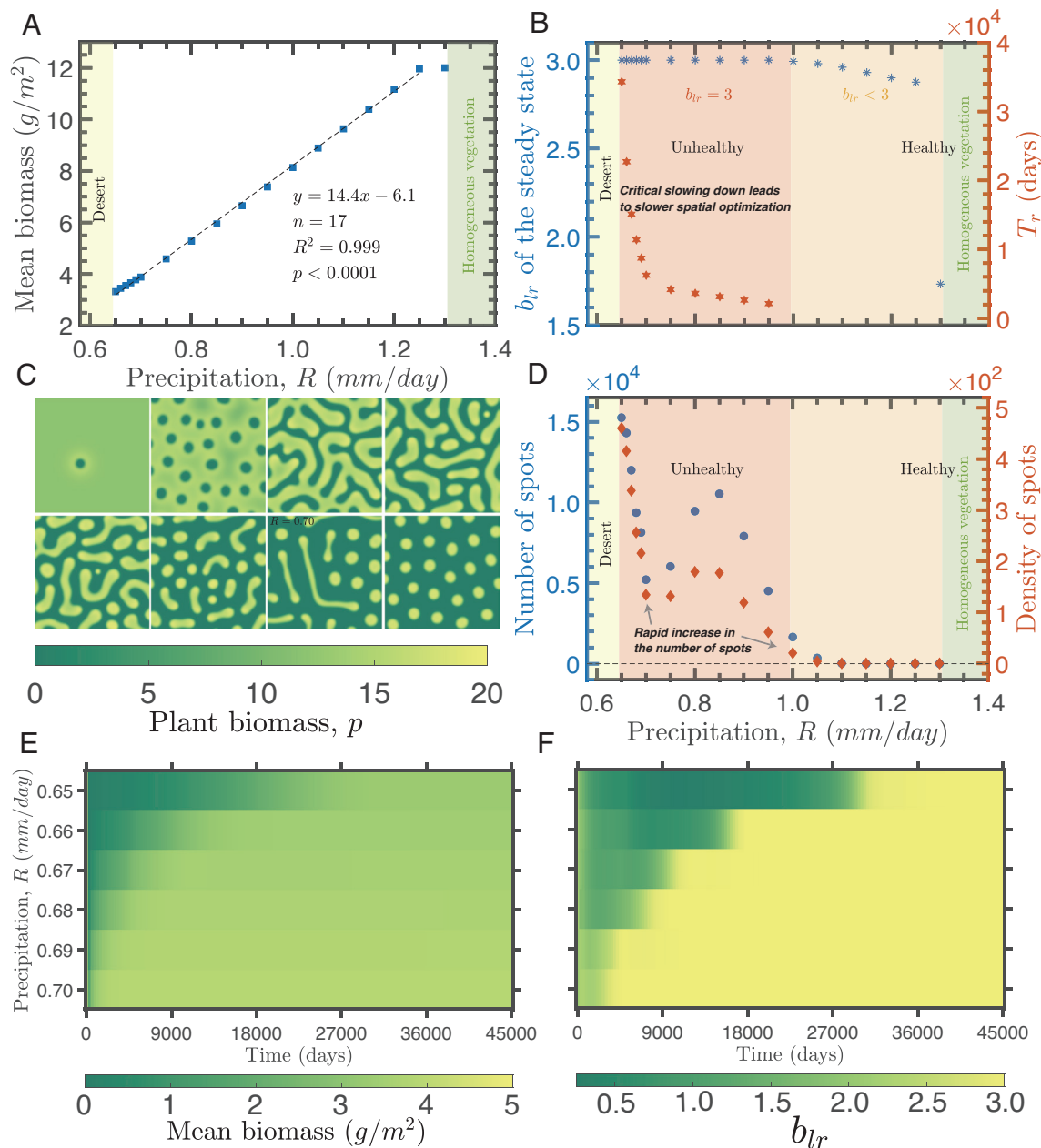
In the early stages of spatial structure evolution, transient vegetation rings can emerge when the rainfall is low (Fig. 4 and *SI Appendix, Figs. S3 and S5*). Detecting transient spatial structures



**Fig. 2.** Spatial optimization of water acquisition in hyperuniform Turing patterns of arid and semi-arid vegetation ecosystems after pattern formation. (A) Schematic representation of Eq. 1 and UPL (Methods). Plants can take up water directly when rain falls on vegetation patches. When rain falls on bare ground, surface water diffuses to the vegetation patches. Since the diffusion rate of surface water is fixed, the length of the water particle movement path (i.e., UPL) determines the water-use efficiency by vegetation. (B) The temporal changes of hyperuniform index and mean biomass. (C) The temporal changes of mean UPL and spatial wavelength. The enhancement of the hyperuniform state leads to a smaller mean UPL. (D) The  $1,000 \times 1,000$  slices from spatial patterns correspond to the black dots in B.

is difficult without purposeful high-temporal-frequency observation. However, many small-scale vegetation rings are easily found worldwide in areas where annual rainfall is less than 200 mm (Fig. 4B). The critical slowing down of spatial structure evolution causes vegetation rings to exist for an extended period and be observable. Meron et al. (37) pointed out that transient vegetation rings are common phenomena in many scale-dependent models of arid vegetation-water ecosystems (37). Inner-patch competition for limited water is the fundamental ecological mechanism for forming transient vegetation rings (51). When a spot vegetation patch is established in the early stage of pattern evolution, it expands outward due to diffusion (Fig. 4A). The plants at the edge of the spot capture all the water transported from the bare ground, while the plants in the center of the spot die because they cannot receive enough water, thus forming a vegetation ring. These vegetation rings then break (please note the broken vegetation ring in Fig. 4B2 and B4), and the broken segments are re-formed into more minor spots or folded stripes with increasing  $b_{lr}$ . Spatial competition for water increases from interpatch to intrapatch, indicating system-wide water scarcity. Transient vegetation rings can represent the critical slowing down of spatial structure evolution and, therefore, serve as a visual early warning spatial signal characterizing an arid vegetation ecosystem close to collapse.

We further examined the evolution of Eq. 1 in the presence of white noise (SI Appendix, Eq. S7) since fluctuations due to unknown elements are essential to account for realistic ecological systems (54). In the presence of white noise, the  $b_{lr}$  of the steady-state Turing patterns can reach 3, although the spatial patterns vary considerably between low and high noise levels (SI Appendix, Fig. S7C–E). The differences in spatial wavelengths at different noise levels are minor, with significant differences only in the early stages of the evolution and almost no differences in the steady states. However, the high noise level dramatically reduces the mean UPL compared to the low noise level, which implies better spatial optimization of water-use efficiency by vegetation in the high-noise scenario (SI Appendix, Fig. S7B). Transient behaviors and steady-state mean biomass of SI Appendix, Eq. S4 are virtually indistinguishable from Eq. 1 at the low noise level. However, SI Appendix, Eq. S4 shows a very different transient behavior and higher steady-state mean biomass from Eq. 1 at the high noise level (SI Appendix, Fig. S7A). In the early stage, both the  $b_{lr}$  and mean biomass rapidly increase to much higher values than those of the low noise level scenario. Then,  $b_{lr}$  slowly evolves toward 3, and the mean biomass reaches a higher value than that of the low noise level scenario. Accompanied by a rapid increase in the  $b_{lr}$  in the high noise level scenario, the transient



**Fig. 3.** The mean biomass, hyperuniform index, and the number of spots in the rainfall parameter space of Eq. 1. (A) In steady states, the linear relationship between mean biomass and rainfall when Turing patterns form. When  $R = 1.3$ , the system forms a local pattern different from the Turing pattern. (B) Hyperuniform index of steady states and  $T_r$  under different rainfall.  $T_r$  represents the time required for vegetation to evolve to a spatial pattern with the hyperuniform index reaching 3.  $T_r$  rises exponentially when the rainfall approaches the tipping point. (C) The 500  $\times$  500 slices from steady spatial patterns under different rainfall. Note that the number of spot vegetation patches increases rapidly with decreasing rainfall. (D) The number/density of spot vegetation patches changes with rainfall. A spot vegetation patch is defined as one with a circularity value exceeding 0.9. The density of spots is defined by the number of spot vegetation patches divided by the mean biomass. (E and F) The temporal changes of mean biomass and hyperuniform index as the ecosystem approaches the tipping point.

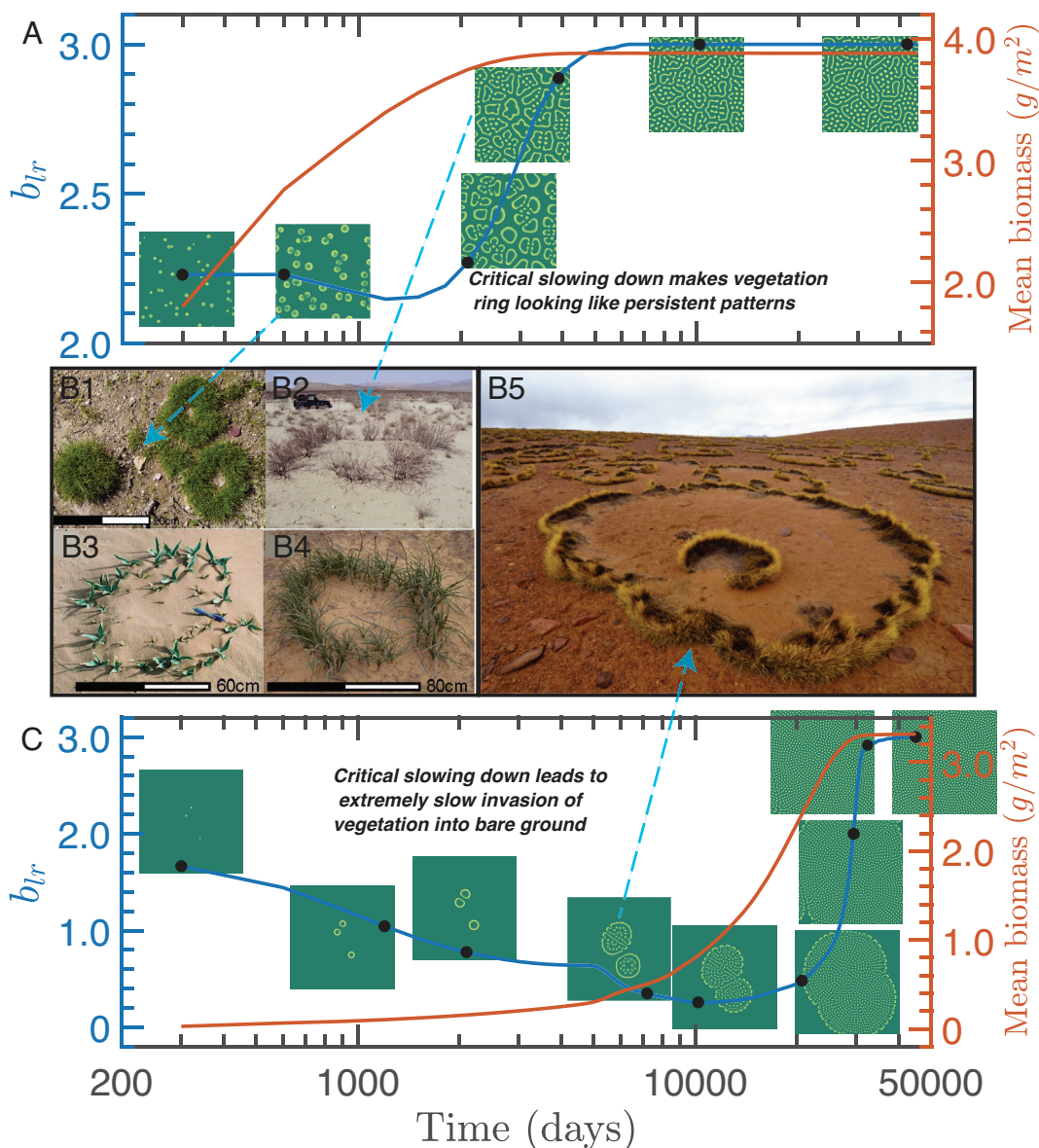
vegetation rings develop in the early stage and then quickly break, and small vegetation spots are the majority in the late stage. Thus, in the presence of white noise, the  $b_{lr}$  still robustly reveals non-negligible transient regimes of this ecosystem.

**Spatial Patterns as a Critical Factor for Maintaining Ecosystem Resilience.** In order to examine the effect of spatial patterns on resilience, we conducted numerical disturbance experiments on these models. Previous studies have established that decreasing rainfall reduces the resilience of steady states of Eq. 1 (38). In this study, we fixed rainfall and focused on how resilience responds to two kinds of disturbances (*Methods*). Specifically, we quantified 1) the resilience of the transient patterns in response to homogeneous

disturbances and 2) the resilience of the steady pattern ( $b_{lr} = 3$ ) in response to both homogeneous and heterogeneous disturbances.

For the first point, we found that recovery time is negatively correlated with  $b_{lr}$  (Fig. 5A and *SI Appendix, Fig. S8A*). A larger  $b_{lr}$  is associated with a higher spatial optimization of water-use efficiency by vegetation. Therefore, a more “mature” spatial pattern has higher resilience in the regime of transient dynamics. Regarding the second point, this ecosystem recovers slowly but can tolerate more intense disturbances under heterogeneous disturbances. In contrast, it recovers quickly but can only tolerate less intense disturbances under homogeneous disturbances (Fig. 5B and C and *SI Appendix, Fig. S8B and C*). While homogeneous disturbances preserve the spatial structure ( $b_{lr}$  does not change after homogeneous disturbances),





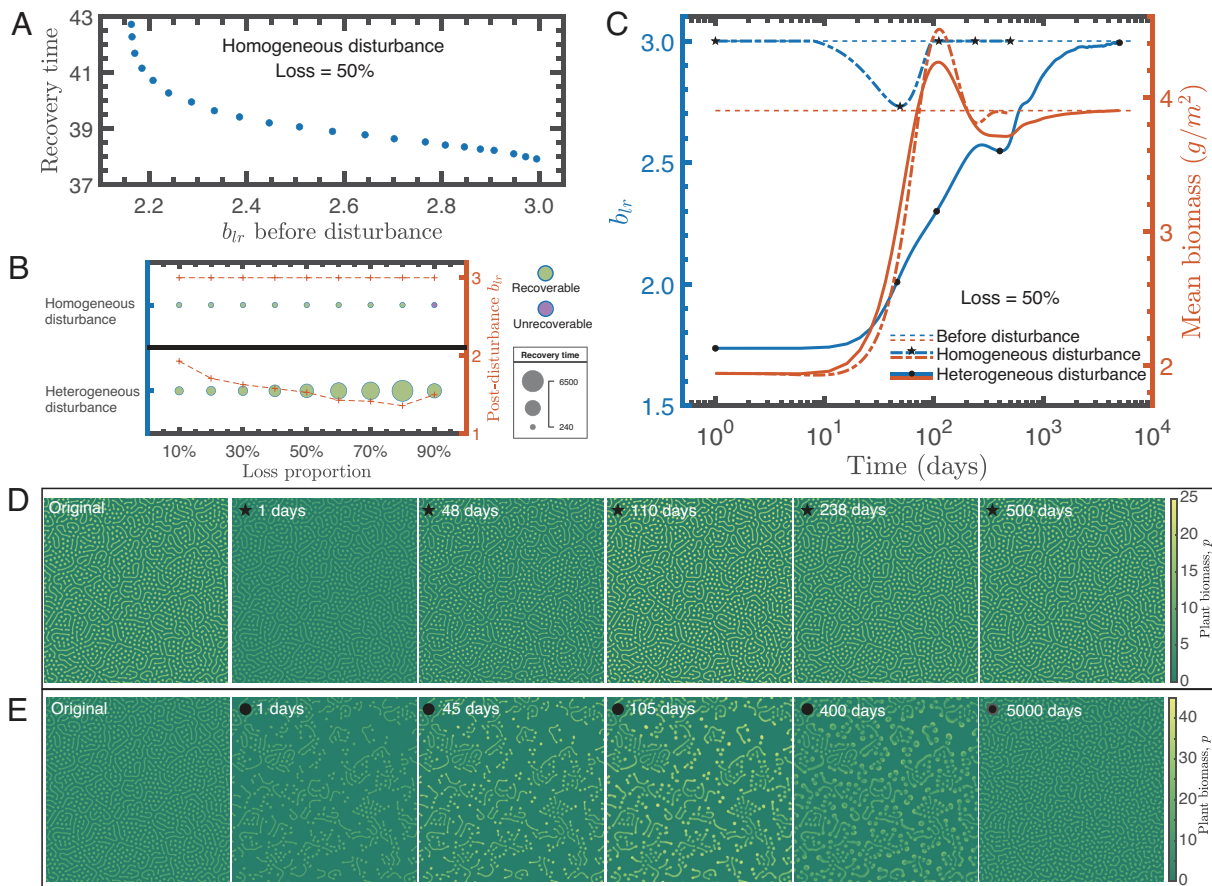
**Fig. 4.** The typical transient behaviors of Turing patterns of arid and semi-arid vegetation ecosystems. (A) The temporal changes of hyperuniform index and mean biomass when  $R = 0.7$ . The *Insets* are  $2,000 \times 2,000$  slices from spatial patterns. (B1–B5) Diverse natural vegetation rings in arid ecosystems. (B1) *Poa bulbosa* L. in Northern Negev, Israel (rainfall 150 mm / y); (B2) *Larrea Tridentata* (DC.) Coville in Lucerne Valley, California, United States (rainfall 98 mm / y); (B3) *Urginea maritima* (L.) Baker in Wadi Rum, Jordan (rainfall 50 mm / y); (B4) *Asphodelus ramosus* L. in the Negev desert, Israel (rainfall 170 mm / y); (B5) *Festuca orthophylla* in Bolivia (rainfall 100 mm / y). B1–B4 are cited from Sheffer et al. (52), and B5 is cited from Chen et al. (53). (C) The temporal changes of hyperuniform index and mean biomass when  $R = 0.65$ . The *Insets* are spatial patterns.

high-intensity disturbances (loss proportion  $\beta_0 = 90\%$  in this case) can push the system to another alternative stable state (i.e., bare ground). From the perspective of dynamical systems, high-intensity homogeneous disturbances can cause the entire system to cross the saddle point to reach the stable state of bare ground.

On the other hand, heterogeneous disturbances destroy the spatial structure ( $b_{lr}$  significantly decreases after heterogeneous disturbances), but they retain some high-biomass vegetation patches. Although high-intensity heterogeneous disturbances cause the spatial mean biomass to cross the saddle point, the biomass of local vegetation patches remains higher than the saddle point. These vegetation patches can recolonize the space during recovery. The resilience of Turing-type arid and semi-arid ecosystems under heterogeneous disturbances depends on the postdisturbance spatial structure rather than the intensity of the

disturbance (Fig. 5B and *SI Appendix*, Fig. S8B). Furthermore, while the recovered pattern is the same as the original pattern under homogeneous disturbance (Fig. 5D), the recovered pattern is different from the original pattern under heterogeneous disturbance (Fig. 5E). This result indicates that homogeneous disturbances do not cause the ecosystem to move away from the original attractor. In contrast, heterogeneous disturbances force the ecosystem to spend more time moving to a new attractor in the solution space.

The resilience of these ecosystems, even under fixed rainfall conditions, is complex and depends on the intensity and spatial structure of disturbance. From the perspective of spatial optimization of water-use efficiency by vegetation, the higher the water-use efficiency, the faster the vegetation will recover. Thus, the larger the  $b_{lr}$  after disturbance, the more resilient the ecosystem is in the recoverable case.



**Fig. 5.** The results of numerical disturbance experiments under fixed rainfall  $R = 0.7$ . We set two kinds of disturbance: homogeneous and heterogeneous (Methods). In brief, homogeneous disturbance keeps the hyperuniform index constant after disturbance. Heterogeneous disturbance reduces hyperuniform index after disturbance. "Recoverable" means the mean biomass can recover to the predisturbance mean biomass. "Unrecoverable" means that vegetation collapse. "Recovery time" is the days to recover a unit of mean biomass. (A) Results for transient vegetation patterns. (B–E) Results for a steady vegetation pattern with mean biomass of  $3.89 \text{ g/m}^2$  and hyperuniform index of 3. (B) The recoverability and recovery time of vegetation under different types and intensities of disturbances. (C) The temporal changes of hyperuniform index and mean biomass in a specific case. (D and E) The spatial patterns corresponding to the symbols in C. "Original" means the predisturbance spatial pattern. The two Original patterns are identical but shown with different color bars. Please note that the recovery time in B is longer than that in A. The recovery in B is the return of both mean biomass and hyperuniform index to the steady state, while the recovery in A is only the return of the mean biomass to the predisturbance state.

## Discussion

The hyperuniform structures found in various systems can optimize certain functions. In materials science, hyperuniformity can enhance transport properties, mechanical stability, and efficient packing of particles, making it promising for applications in optics and photonics (55–57). In biology, hyperuniformity is observed in specific tissues, such as the retina of birds, and can improve visual acuity (58). Through scale-dependent feedback, termite mounds in semi-arid regions form hyperuniform structures called hyperdispersed or overdispersion configurations (24, 25, 59). These studies show that hyperuniform structures enhance multiple ecological functioning of termite mound ecosystems by minimizing the average distance from an arbitrarily chosen point to the nearest termite mound (24). However, during the comparisons between the field landscape and null models with the spatially random distribution of the same number of termite mounds, they only calculated the average distance from a point to the nearest termite mound without quantifying the degree of long-range dispersion. In contrast, by combining density fluctuation and UPL, we confirm that hyperuniform structures resulting from scale-dependent feedback optimize the use of limited resources in ecology. The resource concentration mechanism of Turing-type arid and semi-arid ecosystems involves aggregation due to short-range

facilitation and dispersion due to long-range competition. Spatial competition for limited resources between consumers leads to consumers' arrangement into hyperuniform structures that benefit the efficient use of limited resources. Therefore, scale-dependent feedback is a bottom-up ecological mechanism for vegetation to achieve optimality in arid and semi-arid environments (60). Since Turing patterns are widely present in diverse ecosystems (61), ecosystems with Turing patterns likely optimize ecological functioning through hyperuniform structures.

Competition plays a more significant role than facilitation in evolving Turing patterns of arid and semi-arid ecosystems. In contrast to well-mixed systems represented by ordinary differential equations, partial differential equations introduce more degrees of freedom by incorporating spatial terms, which naturally introduce intraspecific competition between these degrees of freedom. A previous theoretical study has shown that competition between vegetation patches for limited resources leads to repulsive interactions, which accounts for the long-range dispersion of vegetation patterns (62). Intuitively, intraspecific competition among vegetation patches is also reflected in the rapid increase of spots due to decreasing rainfall (Fig. 3D). Although many studies mathematically explain the emergence of spots in scale-dependent models of arid and semi-arid vegetation-water ecosystems (e.g., through the amplitude equation) (37), the ecological significance of the

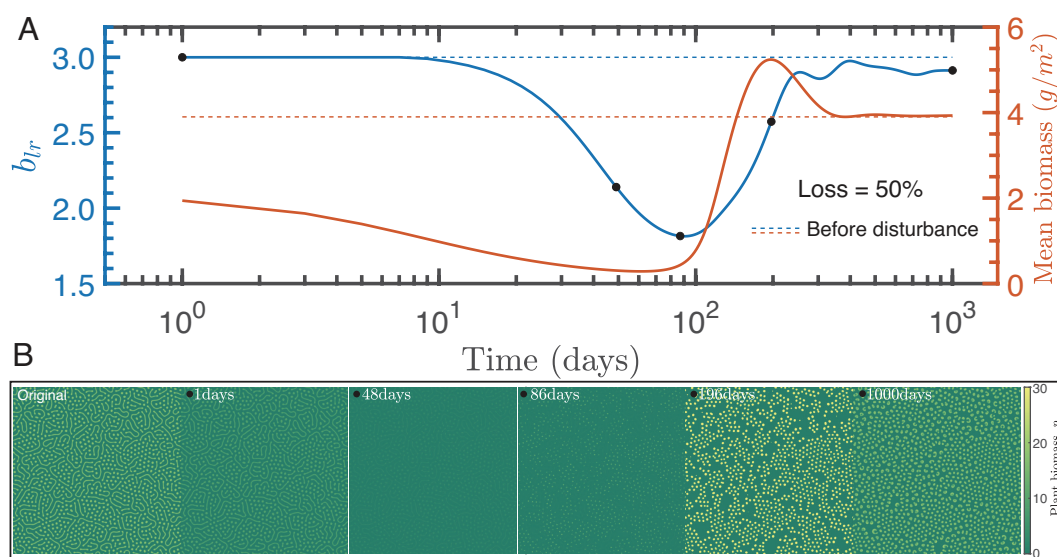


emergence of spots remains ambiguous. On the one hand, surface water can flow to a vegetation patch from all directions since these models describe ecosystems on flat ground. Therefore, a spot is more likely to capture water than a folded stripe because a spot is isotropic, while a folded stripe is anisotropic (5). On the other hand, the water capture efficiency of a spot is weaker than that of a folded stripe since a larger area-perimeter ratio is less conducive to receiving water from the surrounding bare ground. Thus, the labyrinth pattern consisting of large-area folded stripes appears when water is relatively adequate, while small-area spots appear when water is scarce (Fig. 3C). The shift from folded stripes to spots reflects a trade-off of vegetation in adapting to the degraded environment: environmental stress causes vegetation patches to reduce water capture efficiency but increase the likelihood of capturing water. To test this idea, we conducted numerical disturbance experiments of Eq. 1 by applying homogeneous disturbances to vegetation and water, which showed that only the spots could survive, and the folded stripes disappear in the early stage of recovery when water is scarce (Fig. 6). We conclude that the occurrence of spots indicates intense competition for limited water by plants, while water scarcity is approaching limit when all vegetation patches are spots (Figs. 1A1 and A3 and 4C).

Our study reveals important transient properties of ecological Turing patterns in arid and semi-arid ecosystems. Beginning with the seminal work by Lotka and Volterra, stable asymptotic behavior is a central object for understanding the nature of ecosystems (63). Previous studies have mainly focused on the asymptotic state of Turing patterns due to the methodological difficulty in distinguishing transient states from asymptotic states. We propose that the hyperuniform index of the Turing patterns, which is a dimensionless and bounded quantity, can effectively characterize whether the arid and semi-arid ecosystems evolve into an asymptotic state after the appearance of spots (Fig. 3 and SI Appendix, Fig. S6). We further find that when the control parameter of this high-dimensional ecosystem approaches the tipping point, it triggers the long transient dynamics known as “ghosts” (39). These long transient dynamics are characterized by critical slowing down of the spatial structure evolution (Figs. 3 and 4 and SI Appendix, Fig. S6). Decreasing rainfall has been confirmed to cause a decrease

in the resilience of asymptotic arid and semi-arid Turing patterns (38, 64). We further find that with decreasing rainfall, critical slowing down of spatial structure evolution also leads to the loss of resilience of transient states because it slows down the spatial optimization of water-use efficiency by vegetation. The loss of hyperuniform states in natural vegetation Turing patterns indicates a low-resilience transient regime (SI Appendix, Fig. S1). In addition, intense white noise facilitates the  $b_{lr}$  of the ecosystem quickly reaching a high value (SI Appendix, Fig. S7). This implies that the ecosystem can evolve toward a steady state in states of high resilience. In this sense, stronger fluctuations favor the ecosystem in high-resilience states, and this result agrees with the existing perspective that stronger white noise could lead to a better resilience of ecosystems (54, 65). Moreover, transient patterns provide a window to assess resilience; more transient vegetation rings indicate decreasing resilience due to increased competition from interpatches to intrapatch (Fig. 4 and SI Appendix, Figs. S3 and S5). Interestingly, all transient vegetation rings with known formation mechanisms can characterize resilience. For example, transient vegetation rings in salt marshes, induced by the fast-slow dynamics, indicate high resilience (47). In contrast, transient vegetation rings in seagrass beds, induced by the Hopf bifurcation, indicate that the meadow is close to collapse (66). Thus, understanding transient dynamics, including the formation mechanisms and spatial hallmarks of resilience, is necessary for developing effective conservation and management strategies to mitigate the harmful effects of environmental degradation (67).

Quantifying the spatial resilience of ecosystems requires understanding how spatial structure influences ecosystem functioning. Extracting signals from data is essentially information compression (68). Big data and complex neural networks like chatGPT can achieve nearly lossless information compression (69). However, there is limited human access to ecosystem information, which results in the loss of information when extracting spatial resilience metrics. Therefore, spatial resilience metrics must focus on information directly related to ecosystem functioning. Although some spatial early warning signals have been developed (70), their ability to link spatial structure to ecosystem functioning has been questioned (71), making it difficult to assess their robustness (72).



**Fig. 6.** Competition between spots and folded stripes when water is scarce. The rainfall is fixed with  $R = 0.7$ , and we apply homogeneous disturbances to plants, surface water, and soil water in the steady state of Eq. 1. (A) The temporal changes of hyperuniform index and mean biomass after the disturbance. (B) The spatial patterns correspond to the black dots in A. The folded stripes are excluded in 86 d, and the surviving spots reach peak biomass in 196 d. Thus, the folded stripes are “weak competitors” and the spots are “strong competitors.” Subsequently, intraspot competition leads to the emergence of transient vegetation rings (the hyperuniform index still does not reach 3 in 1,000 d).



Nevertheless, these signals can effectively characterize ecosystem resilience if they are linked to ecosystem functioning in some way (73). For Turing-type arid and semi-arid ecosystems, we propose using the hyperuniform index as a metric of spatial resilience after disturbances. Spatial patterns in these ecosystems, shaped by scale-dependent feedback, are critical determinants of ecosystem resilience because they influence the water-use efficiency of vegetation. Spatial patterns are crucial for the recovery rate of forests after intense short-term disturbances (74, 75). In this study, damaging the hyperuniform spatial structure of Turing-type ecosystems through heterogeneous disturbances can lead to a significant increase in the recovery time (Fig. 5B). This implies that overgrazing and wildfire may be more detrimental to the resilience of Turing-type arid and semi-arid ecosystems than flash drought. Moreover, when these arid and semi-arid ecosystems are pushed back into a regime of long transient dynamics, the adverse effects of heterogeneous disturbances can be even more severe. Therefore, the resilience of Turing-type arid and semi-arid ecosystems is dynamic, and the hyperuniform index provides a fresh dynamic perspective for assessing it. In this sense, the time-series data of spatial patterns are necessary to assess the resilience of these fragile ecosystems in the face of a rapidly changing world.

## Methods

**Mathematical Model.** Rietkerk et al. (6) proposed a self-organizing model for arid vegetation-water ecosystems (6). The model treats rainfall as surface water, and surface water can infiltrate into the soil. The infiltration rate of surface water is higher in the vegetation-cover ground and is lower in bare ground. The soil water is partially evaporated and partially converted into plant biomass, and plants die at a fixed rate. Surface water, soil water, and plants diffuse in space. The mathematical expression for this model is shown below:

$$\begin{cases} \frac{\partial O}{\partial t} = R - \alpha \frac{P + k_2 W_0}{P + k_2} O + D_O \Delta O \\ \frac{\partial W}{\partial t} = \alpha \frac{P + k_2 W_0}{P + k_2} O - \frac{g_{\max} W}{W + k_1} P - r_w W + D_W \Delta W, \\ \frac{\partial P}{\partial t} = c \frac{g_{\max} W}{W + k_1} P - dP + D_P \Delta P \end{cases} \quad [1]$$

where,  $P$  ( $\text{g}/\text{m}^2$ ),  $W$  ( $\text{mm}$ ), and  $O$  ( $\text{mm}$ ) are plant biomass, soil water, and surface water, respectively;  $t$  (day) is time;  $\Delta = \frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}$  is Laplacian operator;  $R$  ( $\text{mm}/\text{day}$ ) is rainfall;  $\alpha$  ( $1/\text{day}$ ) is maximum infiltration rate;  $k_2$  ( $\text{g}/\text{m}^2$ ) is a saturation constant of water infiltration;  $W_0$  is a dimensionless scaling factor;  $\alpha W_0$  is water infiltration rate in the bare ground;  $\frac{P + k_2 W_0}{P + k_2}$  means the enhancement of water infiltration by plants;  $g_{\max}$  [ $\text{mm}/(\text{gm}^2 \text{day})$ ] is the maximum soil water uptake by plants;  $k_1$  ( $\text{mm}$ ) is a half-saturation constant;  $\frac{g_{\max} W}{W + k_1} P$  means the ability of plants to acquire soil water increases with soil water, and there is an upper limit  $g_{\max}$ ;  $r_w$  ( $1/\text{day}$ ) is soil water loss due to evaporation;  $c$  [ $\text{g}/(\text{mm} * \text{m}^2)$ ] is a factor describing the efficiency that plants convert water into biomass;  $c \frac{g_{\max} W}{W + k_1} P$  means plants convert water into biomass;  $d$  ( $1/\text{day}$ ) is plant mortality;  $D_O$  ( $\text{m}^2/\text{s}$ ),  $D_W$  ( $\text{m}^2/\text{s}$ ), and  $D_P$  ( $\text{m}^2/\text{s}$ ) are diffusion coefficients for  $O$ ,  $W$ , and  $P$ , respectively, and  $D_O \gg D_W \sim D_P$ ;  $D_O \Delta O$  and  $D_W \Delta W$  represent water movement, and  $D_P \Delta P$  represents plant dispersal by seeds. All parameters are positive, and all variables are non-negative.

**Density Fluctuation.** Density fluctuation can characterize the aggregation or dispersion of the spatial distribution of a system (23, 27, 28):

$$\sigma^2(Q(S)) \sim S^{-b}, \quad [2]$$

where,  $Q(S)$  refers to a density under a specific spatial scale  $S$ ,  $\sigma^2$  is the normalized variance, and  $b$  is the density fluctuation index. For a system in  $D$ -dimensional

Euclidean space, the exponent of density fluctuation,  $b$ , is bounded within the interval of  $[0, D + 1]$ . The spatial Poisson state, the extremely aggregated spatial state, and the extremely dispersive state correspond to exponents of  $D$ ,  $0$ , and  $D + 1$ , respectively. When  $D < b \leq D + 1$ , it is called the hyperuniform state. A system could have distinct short-range density fluctuation  $b_{sr}$  and long-range density fluctuation  $b_{lr}$ , and  $b_{lr}$  is called the hyperuniform index in this study.

Following the idea of measuring density, the density fluctuation of a system can be calculated based on the box-throwing method (SI Appendix, Fig. S9) (76). When a snapshot of an ecosystem (such as a snapshot of Eq. 1) is obtained, we use square boxes of a fixed window size  $S$  and randomly sample over the space on the snapshot. The biomass density  $Q(S)$  within a square box is achieved using the box area to divide the integrated biomass. A dataset of biomass density is obtained by repeating sampling 10,000,000 times. Then, we calculate the variance  $\sigma^2[Q(S)]$  of this dataset for one specific box size. We repeat the above procedure for different window sizes to get a series of  $\sigma^2[Q(S)]$  with different  $S$ . Then, a power law fitting between  $\sigma^2[Q(S)]$  and  $S$  is used for obtaining density fluctuation index  $b$ :

$$\log_{10}\{\sigma^2[Q(S)]\} = -b \log_{10}(S) + \log_{10}(C), \quad [3]$$

where  $C$  is a fitting coefficient related to a specific system. In practice, the maximum box size is usually set to  $1/4$  and  $1/8$  of the system size for simulation and real-world data, respectively (23, 77). The box-throwing method works well for the vector data (such as a point pattern).

Besides the box-throwing method, the calculation of density fluctuation of matrix data can be accelerated via the convolution method executed on the graphics processing unit. A snapshot is convolved with a square matrix of a fixed window size  $S$ , and the elements of the square matrix are  $1/S^2$ . After convolution calculation, we obtain all possible biomass densities. Then, we randomly select 10,000,000 densities from all possible densities to calculate  $\sigma^2[Q(S)]$ . We repeat this procedure for different window sizes to get a series of  $\sigma^2[Q(S)]$  with different  $S$ . The convolution method is merely an accelerated calculation method for the box-throwing method (SI Appendix, Fig. S10). In this research, we apply the convolution method for the model snapshots. For Eq. 1,  $b_{sr}$  and  $b_{lr}$  are calculated within  $S = 0.5 \sim 5(\text{m})$  and  $S = 50 \sim 500(\text{m})$ , respectively.

**Spectral Analysis.** We use the standard spectral analysis method to calculate the spatial wavelength (20, 21). Besides that, we also calculate the circularly averaged spectral density  $\bar{\Psi}(k)$  of a spatial pattern, where  $k$  is the wave number. The circularly averaged spectral density is also called "mean r-spectrum" in ecological and geographic studies (78, 79).

**UPL.** UPL is a hydrological metric commonly used in watershed research (34, 80). It quantifies the distance that a water particle travels before reaching a channel from an unchanneled point. The basic principle underlying UPL is that water flows downhill. When data on a watershed's topography or water level are available, UPL is defined as the steepest descent flow path downslope until the water reaches a channeled pixel. Essentially, UPL provides information about the flow path characteristics within a landscape from a Lagrangian perspective.

In our study, we have access to the spatial distributions of surface water content  $O$  and plant biomass  $P$  from the numerical simulation of Eq. 1. To calculate the UPL, we classify each point in the landscape as either bare ground or vegetation, and consider the former as unchanneled points and the latter as channel points. We assume that the surface water level is proportional to the surface water content  $O$  and use this information to calculate the UPL for each unchanneled point based on the spatial distribution of  $O$ . Finally, we obtain the spatial mean of the UPLs of all bare ground points.

**Two Kinds of Disturbances.** We apply disturbances to a specific ecosystem state by reducing biomass. The predisturbance vegetation state is  $P_1(\vec{x}, t)$ ; the postdisturbance vegetation state is  $P_2(\vec{x}, t)$ ; the loss proportion is  $\beta_0$ ;  $\bar{P}$  is the spatial mean of  $P_1(\vec{x}, t)$ , respectively;  $\{\vec{I}_j, j = 1, 2, \dots, n_0\}$  is the set of vegetation patches in  $P_1(\vec{x}, t)$ .

Homogeneous disturbance:

$$P_2(\vec{x}, t) = (1 - \beta_0) P_1(\vec{x}, t). \quad [4]$$

The hyperuniform indexes of  $P_1(\vec{x}, t)$  and  $P_2(\vec{x}, t)$  are identical under homogeneous disturbance.

Heterogeneous disturbance: we randomly select vegetation patches one by one from  $\{\bar{I}_i\}$  and collect  $m_0$  ( $m_0 < n_0$ ) vegetation patches until the mean biomass of these  $m_0$  vegetation patches equal  $\beta_0 P$ . Then, we set the biomass of these  $m_0$  vegetation patches with 0 to obtain  $P_2(\bar{x}, t)$ . The hyperuniform indexes of  $P_1(\bar{x}, t)$  and  $P_2(\bar{x}, t)$  are different under heterogeneous disturbance unless  $P_1(\bar{x}, t)$  is in a spatial Poisson state (i.e., the density fluctuation index  $b = 2$ ).

**Numerical Simulations.** We numerically simulate Eq. 1 with a square domain  $L^2$  in MATLAB R2022a. We use forward difference and two-order center schemes for the time derivative and Laplace operators, respectively. We use almost the same parameter setting as Rietkerk et al. (6) except  $L = 4096(6)$ . The spatial resolution is 0.5 m, and the temporal resolution is 0.0005 d.  $D_0 = 100$ ,  $D_W = D_p = 0.1$ ,  $\alpha = 0.2$ ,  $W_0 = 0.2$ ,  $r_w = 0.2$ ,  $c = 10$ ,  $g_{\max} = 0.05$ ,  $d = 0.25$ ,  $k_1 = 5$ ,  $k_2 = 5$ . There are two kinds of initial conditions in this research:

- [1] We randomly select 5% grid points and set plant biomass to 100 g / m<sup>2</sup>, and set plant biomass of other grid points to 0; set all grid points to the bare-ground equilibrium of  $W = R/r_w$ ,  $O = R/(\alpha W_0)$ ;
- [2] In numerical disturbance experiments, we apply disturbances to the patterned system states obtained from the previous numerical simulations and then use the disturbed system states as the initial values.

We use periodic boundaries to minimize boundary effects. Simulations are accelerated by the graphics processing units (NVIDIA Tesla K40c), and all details are shown below:

Fig. 1: we use  $R = 0.9$ ,  $t = 3,000$ , and the first initial condition.

Fig. 2: identical to Fig. 1.

Fig. 3: we use the first initial condition. We scan 12 evenly spaced values of  $R$  on the interval  $[0.75, 1.3]$  with  $t = 6,000$ , and scan 6 evenly spaced values of  $R$  on the interval  $[0.65, 0.7]$  with  $t = 45,000$ .

Fig. 4: we use the first initial condition and  $t = 45,000$ . We set  $R = 0.7$  for Fig. 4A and  $R = 0.65$  for Fig. 4C.

Fig. 5: we use the second initial condition and  $R = 0.7$ . For Fig. 5A, we select transient ecosystem states of different hyperuniform indexes from Fig. 4A. Then, we apply disturbance to these ecosystem states and simulate the ecosystem until the mean biomass recovers to predisturbance mean biomass. For Fig. 5 B–E, we select the final state of the ecosystem from Fig. 4A and then apply disturbances to the vegetation state and simulate the ecosystem until the mean biomass and the hyperuniform index recover to predisturbance mean biomass and 3, respectively.

Fig. 6: we use the second initial condition,  $R = 0.7$ , and  $t = 1,000$ . We select the final state of the ecosystem from Fig. 4A and then apply disturbances to the ecosystem state.

**Analysis of Field Vegetation Patterns.** We select five areas from the literature studying vegetation Turing patterns in arid and semi-arid ecosystems (15, 19, 40, 73, 81). These areas have no significant topographic changes, such as slopes, mountains, canyons, rivers, or highways. For the vegetation patterns shown in Fig. 1 A1–A4, we take the screenshots (RGB images) from Google Earth Pro and extract the vegetation from the images in ENVI 5.3 software. We perform image enhancement: by applying a median filter and histogram equalization. Then, we use a supervised classification method (random forest) to classify the vegetation pixels as “1” and the other pixels as “0.” Subsequently, the density fluctuation of the binary images is calculated by the convolution method. For the vegetation patterns shown in *SI Appendix, Fig. S2*, we take the screenshots (RGB images) from Barbier et al. (19). We transform RGB images to gray-scale images. Since Barbier et al. (19) stated that the images' gray-scale values can be considered a monotonic function of phytomass, the density fluctuation of the gray-scale images is directly calculated by the convolution method.

**Data, Materials, and Software Availability.** The codes to reproduce the results of this study are available on the publicly available repository GitHub (82) [https://github.com/liufengyinxue/Hidden\\_order\\_arid\\_Turing](https://github.com/liufengyinxue/Hidden_order_arid_Turing). All study data are included in the article and *SI Appendix*.

After the release of this work, Quan-Xing Liu from Shanghai Jiao Tong University informed me that two other independent works of hyperuniform states of ecological and geological systems were in preparation, respectively (83, 84).

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